SECTIONS OF THE PULLBACK SHEAF ON THE LIMIT OF A SPECTRAL SYSTEM COMPUTE AS THE COLIMIT OF PULLBACKS

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Allowed standard prerequisites (editorial).

- Topology, Lemma topology-lemma-inverse-limit-spectral-spaces-nonempty (canonical tag 0A2W): a directed inverse limit of nonempty closed subsets of spectral spaces is nonempty. cited at proof lines 3804-3806. Yields a point where the two sections differ in the limit, forcing some Z_k empty.
- Topology, Lemma topology-lemma-directed-inverse-limit-spectral-spaces (canonical tag 0A2Z): the limit of a directed system of spectral spaces is spectral and projections behave accordingly. cited at proof lines 3814-3816. Ensures p_i^{-1}(U_i) is quasi-compact open in the spectral space X.
- Topology, Lemma topology-lemma-descend-opens (canonical tag 0A30): coverings of the limit descend along some transition map of a directed spectral system. cited at proof lines 3828-3830. Descends the finite covering of p_i^{-1}(U_i) to a covering of f_a^{-1}(U_i) for some a : j -> i.

Source reference. Original lemma 0A2W: sources/stacks-chow/tags.txt, source lines 12719–12719. Cited at proof lines 3804-3806 for injectivity; load-bearing inside the counted span.

Source reference. Original lemma 0A2Z: sources/stacks-chow/tags.txt, source lines 12722–12722. Cited at proof lines 3814-3816 to ensure quasi-compactness; load-bearing inside the counted span.

Source reference. Original lemma 0A30: sources/stacks-chow/tags.txt, source lines 12723–12723. Cited at proof lines 3828-3830 to descend the covering; load-bearing inside the counted span.

Lemma 0.2. *In the situation described above, let $i \in \mathrm{Ob}(\mathcal{I})$ and let $\mathcal{G}$ be a sheaf on X_i. For $U_i \subset X_i$ quasi-compact open we have*

$$p_i^{-1}\mathcal{G}(p_i^{-1}(U_i)) = \mathrm{colim}_{a:j\to i} f_a^{-1}\mathcal{G}(f_a^{-1}(U_i))$$

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Proof. Let us prove the canonical map $\mathrm{colim}_{a:j\to i} f_a^{-1}\mathcal{G}(f_a^{-1}(U_i)) \to p_i^{-1}\mathcal{G}(p_i^{-1}(U_i))$ is injective. Let s, s' be sections of $f_a^{-1}\mathcal{G}$ over $f_a^{-1}(U_i)$ for some $a : j \to i$. For $b : k \to j$ let $Z_k \subset f_{a\circ b}^{-1}(U_i)$ be the closed subset of points x such that the image of s and s' in the stalk $(f_{a\circ b}^{-1}\mathcal{G})_x$ are different. If Z_k is nonempty for all $b : k \to j$, then by Topology, Lemma 0A2W we see that $\lim_{b:k\to j} Z_k$ is nonempty too. Then for $x \in \lim_{b:k\to j} Z_k \subset X$ (observe that $\mathcal{I}/j \to \mathcal{I}$ is initial) we see that the image of s and s' in the stalk of $p_i^{-1}\mathcal{G}$ at x are different too since $(p_i^{-1}\mathcal{G})_x = (f_{b\circ a}^{-1}\mathcal{G})_{p_k(x)}$ for all $b : k \to j$ as above. Thus if the images of s and s' in $p_i^{-1}\mathcal{G}(p_i^{-1}(U_i))$ are the same, then Z_k is empty for some $b : k \to j$. This proves injectivity.

Surjectivity. Let s be a section of $p_i^{-1}\mathcal{G}$ over $p_i^{-1}(U_i)$. By Topology, Lemma 0A2Z the set $p_i^{-1}(U_i)$ is a quasi-compact open of the spectral space X. By construction of the pullback sheaf, we can find an open covering $p_i^{-1}(U_i) = \bigcup_{l\in L} W_l$, opens $V_{l,i} \subset X_i$, sections $s_{l,i} \in \mathcal{G}(V_{l,i})$ such that $p_i(W_l) \subset V_{l,i}$ and $p_i^{-1}s_{l,i}|_{W_l} = s|_{W_l}$. Because X and X_i are spectral and $p_i^{-1}(U_i)$ is quasi-compact open, we may assume L is finite and W_l and $V_{l,i}$ quasi-compact open for all l. Then we can apply Topology, Lemma 0A30 to find $a : j \to i$ and open covering $f_a^{-1}(U_i) = \bigcup_{l\in L} W_{l,j}$ by quasi-compact opens whose pullback to X is the covering $p_i^{-1}(U_i) = \bigcup_{l\in L} W_l$ and such that moreover $W_{l,j} \subset f_a^{-1}(V_{l,i})$. Write $s_{l,j}$ the restriction of the pullback of $s_{l,i}$ by f_a to $W_{l,j}$. Then we see that $s_{l,j}$ and $s_{l',j}$ restrict to elements of $(f_a^{-1}\mathcal{G})(W_{l,j}\cap W_{l',j})$ which pullback to the same element $(p_i^{-1}\mathcal{G})(W_l \cap W_{l'})$, namely, the restriction of s. Hence by injectivity, we can find $b : k \to j$ such that the sections $f_b^{-1}s_{l,j}$ glue to a section over $f_{a\circ b}^{-1}(U_i)$ as desired. □